

Basic details

Time: MWF from 1:50pm–2:45pm

Location: Purnell Hall 236

Instructor: Jacob Calvert

- Email: calvert@udel.edu
- Office: Ewing Hall 509

Contact: The best way to reach me is by Canvas message. If you email me instead, please begin the subject line with “MATH 637”

Office Hours: MW from 2:45pm–3:45pm in Ewing Hall 509

Web page: <https://jacobcalvert.com/teaching/math637fall2026/>

- Lectures, homework assignments, and code will be posted to this website. We will use Canvas for assignment deadlines and grades.

Course overview

This course will introduce fundamental techniques for data exploration, prediction, and inference. Our main objective will be to develop a mathematical understanding of data science techniques—their assumptions, properties, and limitations—alongside their practical implementation and use. Several of our meetings will be dedicated to problem-solving and programming sessions. Final projects will evaluate both an understanding of course material as well as the critical thinking skills at the core of effective data science.

Representative list of topics:

- the data science life cycle
- dimension reduction (e.g., singular value decomposition, principal component analysis, collaborative filtering, autoencoders)
- clustering and density estimation (e.g., k -means algorithm, Gaussian mixture models, expectation-maximization algorithm, variational autoencoders)
- regression (e.g., least squares, maximum likelihood and maximum a posteriori estimation, ridge and lasso regularization, Bayesian linear regression)
- classification (e.g., k -nearest neighbors, logistic regression, decision trees, random forests, gradient boosting, support vector machines)
- neural networks (e.g., multilayer perceptrons, backpropagation, stochastic gradient descent)
- model assessment and uncertainty quantification (e.g., cross-validation, calibration, conformal prediction)
- explainability (e.g., generalized additive models, explainable boosting machines, Rashomon effect)
- frontier (e.g., tabular foundation models)

Prerequisites:

- Probability theory and basic statistics at the level of MATH 350
- Multivariable calculus at the level of MATH 243
- Linear algebra at the level of MATH 349
- Basic programming skills, ideally in Python

Learning outcomes

By the end of this course, you should be able to:

- define, analyze, and implement standard techniques for data exploration, prediction, and inference;
- critically evaluate techniques for a given data science task on the basis of their underlying assumptions, mathematical properties, and the task's demands;
- thoughtfully interpret and effectively communicate the results of data exploration and inference;
- learn new data science techniques and programming libraries from research papers and online resources; and
- responsibly incorporate AI models into the data science workflow.

Resources

There is no one textbook that covers all the material that we will discuss. However, the closest in terms of presentation of the core topics is:

- *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong (Cambridge University Press, 2020), which is freely available at: <https://mml-book.github.io/>

We will also draw on several other freely-available textbooks:

- *An Introduction to Statistical Learning with Applications in Python* by Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (Springer, 2023)
Available at: <https://www.statlearning.com/>
- *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* by Trevor Hastie, Robert Tibshirani, and Jerome Friedman (Springer; 2nd edition, 2009)
Available at: <https://hastie.su.domains/ElemStatLearn/>
- *Deep Learning: Foundations and Concepts* by Christopher M. Bishop with Hugh Bishop (Springer, 2024)
Available at: <https://www.bishopbook.com/>
- *Veridical Data Science: The Practice of Responsible Data Analysis and Decision Making* by Bin Yu and Rebecca L. Barter (MIT Press, 2024)
Available at: <https://vdsbook.com/>

Assessments

We will periodically have short in-class quizzes to assess basic ideas and techniques from recent lectures or assignments. We will regularly use our meetings to solve problems and program techniques from the class together; active participation in these sessions is expected. Homework will often be an extension of the work we begin in these sessions, and quiz and midterm questions will draw from both. We will have an in-class midterm to assess your understanding of the basic techniques of each

main type. The final project, which will be done individually or in small groups, will involve both a written report and a presentation with Q&A.

- In-class quizzes: 15%
- Homework: 25%
- Midterm exam: 30%
- Final project report: 15%
- Final project presentation, Q&A: 15%

In-class quizzes. Short quizzes will be given regularly. There will be no make-up quizzes, but the three lowest scores will be dropped to account for this.

Homework. There will be ≈ 10 assignments over the semester. Late homework will generally not be accepted, but you will have two 72-hour exceptions during the semester, provided that you tell me you are using it before the deadline has passed. Additionally, the two lowest homework scores will be dropped.

Use of AI. The use of AI is permitted only as a programming aid, and only on the homework and the final project. It is not to be used for solving theory problems, for any of the prose you write in the course, for explanations of your reasoning, or for judgment calls about modeling. Any permitted use of AI must be explicitly declared and explained: End the submission with a short note naming the tool and describing exactly how you used it. In all cases, you are responsible for what you submit.

Final project. Rather than fix a group size in advance, we will discuss together what scope of project is reasonable for the number of members. Your final project grade will partly reflect your use of the more advanced techniques presented after the midterm. The assessment will depend heavily on how well you can answer questions about the choices you made at each step of the data science process, and how well you justify the conclusions you drew from it. This applies to the report, but especially to the Q&A following your presentation. A brief proposal and a progress check will be due earlier in the semester; dates will be posted on the course web page.

Grading scale

We will use the default Canvas grading scale of: $A \geq 94\%$, $A- \geq 90\%$, $B+ \geq 87\%$, $B \geq 84\%$, $B- \geq 80\%$, $C+ \geq 77\%$, $C \geq 74\%$, $C- \geq 70\%$, $D+ \geq 67\%$, $D \geq 64\%$, $D- \geq 61\%$, $F < 61\%$. Any changes will be to the students' benefit.

Platforms

- We will use Python via Google Colab (<https://colab.research.google.com>) for coding and programming assignments.
- We will use LaTeX via Overleaf (<https://www.overleaf.com>) to write the final project report.

Academic Integrity

Please familiarize yourself with UD policies regarding academic dishonesty. To falsify the results of one's research, to steal the words or ideas of another, to cheat on an assignment, to re-submit the same assignment for different classes, or to allow or assist another to commit these acts corrupts the educational process. Students are expected to do their own work and neither give nor receive unauthorized assistance. Complete details of the university's academic integrity policies and procedures can be found at <https://sites.udel.edu/studentconduct/sgup/>

Office of Student Conduct, 218 Hullihen Hall, (302) 831-2117. E-mail: student-conduct@udel.edu

In particular, note that:

- Copying solutions in whole or in part from other students or any other source without acknowledgement constitutes cheating.
- Any student found cheating risks automatically failing the class and will be referred to the Office of Student Conduct.
- You can discuss with other students, but must write up your own solutions/codes.
- Please note your collaborators on your submissions.

UD Policies

Harassment and Discrimination. The University of Delaware works to promote an academic and work environment that is free from all forms of discrimination, including harassment. As a member of the community, your rights, resource and responsibilities are reflected in the non-discrimination and sexual misconduct policies. Please familiarize yourself with these policies at <https://www.udel.edu/oei>

You can report any concerns to the University's Office of Equity and Inclusion, at 305 Hullihen Hall, (302) 831-8063 or you can report anonymously through UD Police (302) 831-2222 or the EthicsPoint Compliance Hotline at <https://www1.udel.edu/compliance>

You can also report any violation of UD policy on harassment, discrimination, or abuse of any person at this site: <https://sites.udel.edu/sexualmisconduct/how-to-report/>

Faculty Statement on Disclosures of Instances of Sexual Misconduct. If, at any time during this course, I happen to be made aware that a student may have been the victim of sexual misconduct (including sexual harassment, sexual violence, domestic/dating violence, or stalking), I am obligated by federal law to inform the university's Title IX Coordinator. The university needs to know information about such incidents to, not only offer resources but to ensure a safe campus environment. The Title IX Coordinator will decide if the incident should be examined further. If such a situation is disclosed to me in class, in a paper assignment, or in office hours, I promise to protect your privacy—I will not disclose the incident to anyone but the Title IX Coordinator.

For more information on Sexual Misconduct policies, where to get help, and reporting information, please refer to <https://www.udel.edu/sexualmisconduct>

At UD, we provide 24/7/365 crisis assistance and victim advocacy and counseling. Contact 302-831-1001 to get in touch with a sexual offense support advocate, as well as confidential and anonymous counseling services for other concerns.

Accommodations for Students with Disabilities. Any student who thinks he/she may need an accommodation based on a disability should contact the Office of Disability Support Services (DSS) office as soon as possible. Students who have documentation of their need for accommodation should register via the SAM platform: <https://andes.accessiblelearning.com/UDEL/>

Reach DSS in the following ways: Phone: 302-831-4643; Email: dssoffice@udel.edu or visit at 240 Academy Street, Alison Hall Suite 130.

Non-Discrimination. The University of Delaware does not discriminate against any person on the basis of race, color, national origin, sex, gender identity or expression, sexual orientation, genetic information, marital status, disability, religion, age, veteran status or any other characteristic protected by applicable law in its employment, educational programs and activities, admissions policies,

and scholarship and loan programs as required by Title IX of the Educational Amendments of 1972, the Americans with Disabilities Act of 1990, Section 504 of the Rehabilitation Act of 1973, Title VII of the Civil Rights Act of 1964, and other applicable statutes and University policies. The University of Delaware also prohibits unlawful harassment including sexual harassment and sexual violence.

For inquiries or complaints related to non-discrimination policies, please contact: Office of Equity and Inclusion, Email: oei@udel.edu; Phone: (302) 831-8063, 305 Hullihen Hall Newark, DE 19716

For complaints related to Section 504 of the Rehabilitation Act of 1973 and/or the Americans with Disabilities Act, please contact: Phone: 302-831-4643; Email: dssoffice@udel.edu or visit at 240 Academy Street, Alison Hall Suite 130, or contact the U.S. Department of Education - Office for Civil Rights.